

Mark schemes

Q1.

	Marking Instructions	AO	Marks	Typical Solution
(a)	Translates proportionality into a differential equation involving $\frac{dx}{dt}$, t , x and a constant of proportionality.	AO3.3	B1	$\frac{dx}{dt} = \frac{k(8-t)}{x}$
	Substitutes $t = 2$, $x = 336$, $\frac{dx}{dt} = 72$ to find k	AO1.1a	M1	$72 = \frac{k(8-2)}{336}$ $k = 4032$
	Obtains correct value of k shows given result AG	AO2.1	R1	$\frac{dx}{dt} = \frac{4032(8-t)}{x}$ $x \frac{dx}{dt} = 4032(8-t)$
(b)	Integrates one side correctly	AO1.1a	M1	$\int x dx = \int 4032(8-t) dt$
	Integrates both sides correctly condone missing c	AO1.1b	A1	$\frac{1}{2}x^2 = 4032\left(8t - \frac{t^2}{2}\right) + c$
	Uses conditions to show $c = 0$ and correctly obtains given result. AG	AO2.1	R1	$\frac{1}{2} \times 336^2 = 4032\left(8 \times 2 - \frac{2^2}{2}\right) + c$ $c = 0$ $x^2 = 64512t - 4032t^2$ $\therefore x^2 = 4032t(16-t)$
(c) (i)	Translates rate of growth into $\frac{dx}{dt} = 24$ and uses in model for rate of sales	AO3.3	M1	$24x = 4032(8-t)$ $x = 168(8-t)$ $(168(8-t))^2 = 4032t(16-t)$
	Eliminates x to form quadratic equation or inequality in t	AO3.1a	M1	$t^2 - 16t + 56 = 0$ $t = 5.171\ldots$ or $10.828\ldots$ 5 hours 10 minutes
	Obtains correct equation or inequality in t Need not be simplified	AO1.1b	A1	Earliest time 14:40
	Obtains $t = 5.17$	AO1.1b	A1	
	Converts their t into hours and minutes	AO3.4	A1	



	Interprets the closing time as 14:40 09:30 + their converted time	AO3.2a	R1F	
(ii)	Explains in context that when the stall opens there will be zero sales	AO3.5a	E1	When the stall opens there are zero sales
	Explains that when $x = 0$ the model is undefined	AO3.5a	E1	When $x = 0$ $\frac{dx}{dt}$ is undefined as the denominator is zero
Total 14 marks				

Q2.

	Marking Instructions	AO	Marks	Typical Solution
(a)	Translates rate of change into $\frac{dm}{dt}$	AO3.3	M1	$\frac{dm}{dt} = -\frac{k}{\sqrt[3]{m}}$
	Translates inverse proportionality by using $\frac{1}{\sqrt[3]{m}}$ in an equation (no need to see minus sign or k to earn this mark)	AO3.3	M1	
	Forms correct equation with correct notation $\frac{dm}{dt} = -\frac{k}{\sqrt[3]{m}}$ or equivalent eg $-\frac{dm}{dt} = \frac{k}{\sqrt[3]{m}}$ or $\frac{dm}{dt} = -km^{-\frac{1}{3}}$	AO1.1b	A1	
(b)	Gives a relevant criticism of the assumption	AO3.5b	E1	Sam's mass is unlikely to follow this model all the time, when he eats his mass will go up. OR Sam's assumption predicts that his mass will decrease indefinitely.

Total 4 marks

Q3.

	Marking Instructions	AO	Marks	Typical Solution
(a)	Translates proportionality into a differential equation involving $\frac{dA}{dt}$, A and a constant of proportionality.	AO3.3	M1	$\frac{dA}{dt} \propto A$ $\Rightarrow \frac{dA}{dt} = kA$ $\Rightarrow \int \frac{1}{A} dA = \int k dt$
	Separates variables	AO1.1a	M1	$\Rightarrow \ln A = kt + c$ $\Rightarrow A = e^{kt+c}$
	Integrates both of 'their' sides correctly	AO1.1b	A1F	$\Rightarrow A = Be^{kt}$ AG
	Constructs a rigorous mathematical argument that supports use of the given model. AG Only award if they have a completely correct solution, which is clear, easy to follow and contains no slips.	AO2.1	R1	
(b)	States correct value of B	AO1.1b	B1	$B = 0.25$ or $B = \frac{1}{4}$
(i)				
(ii)	Uses $t = 20$ and $A = 0.5$ to find k	AO3.1b	M1	When $t = 20$, $A = 0.5$ $\Rightarrow 0.5 = 0.25e^{20k}$
	Finds correct value of k	AO1.1b	A1	$\Rightarrow 20k = \ln 2$ $\Rightarrow k = \frac{1}{20} \ln 2$
	Substitutes 'their' k to get A in terms of t	AO1.1a	M1	$\Rightarrow A = \frac{1}{4} (e^{\ln 2})^{\frac{t}{20}}$
	Constructs rigorous and convincing argument to show $A = 2^{\frac{t}{20}-2}$ Using correct notation throughout. AG	AO2.1	R1	$\Rightarrow A = 2^{-2} \times 2^{\frac{t}{20}}$ $A = 2^{\frac{t}{20}-2}$ AG
(iii)	Uses the model to set up correct equation and	AO3.4	M1	$2\pi = 2^{\frac{t}{20}-2}$

attempt to find t			$t = 93.03$ days
Finds correct value of t	AO1.1b	A1	

(c)	States any sensible and relevant limitation of the model that is specified in terms of the pond, area, weed, rate of change or time.	AO3.5b	E1	Model predicts that the area of weed will increase without limit and this is not possible since the area of the pond is 4π
(d)	Any sensible and relevant refinement to the model that is specified in terms of the pond, area, weed, rate of change or time	AO3.5c	E1	Introduce a limiting factor such as fish eating weed or rate of growth decreases as surface area covered
Total 13 marks				

Q4.

$$\frac{dh}{dt}$$

$$\frac{dh}{dt} \text{ seen}$$

B1

1

$$a = 1.3 \text{ or } a = -1.3$$

B1

1

$$k = \frac{\pi}{6} \text{ or } k = \frac{2\pi}{12}$$

B1

1

[3]

Q5.

(a) $\frac{dh}{dt}$

B1

$$\text{derivative} = * \times (2 - h)$$

Use of $2 - h$ or $h - 2$;

* is a constant or expression in h and / or t .

M1

$$\frac{dh}{dt} = k(2 - h)$$

All correct; must be (2 - h)

A1

3

(b) (i) $\int x\sqrt{2x-1} \, dx = \int \frac{1}{15} dt$

*Correct separation and notation;
condone missing integral signs.*

B1

$$= \frac{1}{15} t$$

B1

Substitute $u = 2x - 1$

$$\int x\sqrt{2x-1} \, dx = \int \frac{1}{2}(u+1)\sqrt{u} \frac{1}{2} du$$

*Suitable substitution and attempt to write integral in terms
of u including dx replaced by $\frac{1}{2} du$ or $2du$.*

M1

$$= \left(\frac{1}{4}\right) \int u^{\frac{3}{2}} + u^{\frac{1}{2}} du$$

$\frac{1}{4}$ need not be seen

A1

$$= \frac{1}{4} \left(\frac{2}{5} u^{\frac{5}{2}} + \frac{2}{3} u^{\frac{3}{2}} \right) (+ C)$$

Integration correct including $\frac{1}{4}$

A1

$$x = 1; t = 0$$

$$u = 1, t = 0 \quad \frac{1}{4} \left(\frac{2}{5} + \frac{2}{3} \right) + C = 0$$

*Use $x = 1, t = 0$ to find a value for constant C from
equation in x and t .*

M1

$$C = -\frac{4}{15}$$

$$C = -0.2666\dots$$

$$C = -0.267 \text{ or better}$$

A1

$$t = \frac{1}{2} \left(3(2x-1)^{\frac{5}{2}} + 5(2x-1)^{\frac{3}{2}} \right) - 4$$

$$\text{ISW } t = (2x-1)^{\frac{3}{2}} (3x+1) - 4$$

A1

Alternative (Parts)

As before

(B1B1)

$$u = x, \frac{dv}{dx} = (2x-1)^{\frac{1}{2}}$$

Attempt to use parts

(M1)

$$du = 1 \quad v = k(2x-1)^{\frac{3}{2}}$$

$$\int x\sqrt{2x-1} \, dx = x \frac{1}{3}(2x-1)^{\frac{3}{2}} - \int \frac{1}{3}(2x-1)^{\frac{3}{2}} \, dx$$

Condone missing dx

(A1)

$$= x \frac{1}{3}(2x-1)^{\frac{3}{2}} - \frac{1}{15}(2x-1)^{\frac{5}{2}} (+C)$$

(A1)

$$x = 1, t = 0 \quad \frac{1}{3} - \frac{1}{15} + C = 0$$

Use $x = 1$, $t = 0$ to find a value for constant C from equation in x and t

(M1)

$$C = -\frac{4}{15}$$

$$C = -0.2666\dots$$

$$C = -0.267 \text{ or better}$$

(A1)

$$t = 5x(2x-1)^{\frac{3}{2}} - (2x-1)^{\frac{5}{2}} - 4$$

$$\text{ISW } t = (2x-1)^{\frac{3}{2}}(3x+1) - 4$$

(A1)

8

(ii) $x = 2 \quad t = 32.4$ (minutes)
 32.4 or better (32.373...)

B1

1

[12]

Q6.

(a) $\frac{dA}{dt}$

$$= -k$$

B1

B1

2

(b) (i) $A = -kt (+ C)$
 Integrate

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M1

$$C = 4\pi \times 60^2$$

$$C \text{ correct from } A = \pm k + C$$

A1

$$4\pi \times 30^2 = -9k + 4\pi \times 60^2$$

Use $r = 30 \quad t = 9$ and attempt to find k ,
 as far as $k = \dots$

$$k = 1200\pi$$

m1



$$\begin{aligned}
 A &= -1200\pi t + 14400\pi \\
 &= 1200\pi(12 - t) \\
 &\quad \text{AG CSO}
 \end{aligned}$$

A1

4

(ii) $t = 12$ (days)

B1

1

[7]

Q7.

(a) (i) $t = 0 \quad h = A(1 - 1) = 0$

B1

1

(ii) $57 = A \left(1 - e^{-\frac{12}{4}} \right)$

M1

$$A = \frac{57}{(1 - e^{-1})} \approx 60$$

Or 59.9... seen.

$A = \text{correct expression} \approx 60 \text{ 2 sf}$

A1

2

(b) (i) $h = 48 \quad \frac{48}{60} = 1 - e^{-\frac{1}{4}t}$

M1

$$\ln \left(e^{-\frac{1}{4}t} \right) = \ln \left(\frac{1}{5} \right)$$

m1

$$-\frac{1}{4}t = -\ln 5 \Rightarrow t = 4 \ln 5$$

A1

3



$$(ii) \quad \frac{dh}{dt} = -\frac{1}{4} \times -60 \times e^{-\frac{1}{4}t}$$

Differentiate, condone sign errors

M1

$$60e^{-\frac{1}{4}t} = 60 - h \Rightarrow \frac{dh}{dt} = \frac{1}{4}(60 - h)$$

Eliminate $e^{-\frac{1}{4}t}$

m1

$$\frac{dh}{dt} = 15 - \frac{h}{4}$$

CSO, AG

A1

3

$$(iii) \quad h = 8$$

B1

1

[10]

Q8.

$$(a) \quad \int \frac{dx}{\sqrt{x+1}} = \int -\frac{1}{5} dx$$

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*Correct separation; or $\frac{dt}{dx} = -5(x+1)^{\frac{1}{2}}$
Condone missing integral signs*

B1

$$2\sqrt{x+1} = -\frac{1}{5}t \quad (+C)$$

Correct integrals; condone $\frac{\sqrt{x+1}}{\frac{1}{2}}$

B1B1

$$x = 80 \quad t = 0 \quad C = 2\sqrt{81}$$

Use (0, 80) to find a constant C

M1

$= 18$

F on integrals if in form $\sqrt{x+1} = at + c$

A1F

$$x = \left(9 - \frac{1}{10}t\right)^2 - 1$$

OE; CSO; x = correct expression in t

A1

6

(b) $t = 60 \quad x = f(60)$

Evaluate $f(60)$, ie $x = \dots$ (C not required)

M1

$= 8$

CSO

A1

2

(c) (i) $\frac{dA}{dt} = kA(9 - A)$

$$\frac{dA}{dt} = \text{product involving } A; k \text{ required}$$

Condone terms in t

M1A1

2

(ii) $4.5 = \frac{9}{1 + 4e^{-0.09t}}$

Condone one slip in denominator

M1

$$e^{-0.09t} = \frac{1}{4}$$

A1

$$-0.09t = \ln\left(\frac{1}{4}\right)$$

Take $\ln$ correctly

m1

$$t = \frac{\ln\left(\frac{1}{4}\right)}{-0.09}$$

$$= 15.4 \text{ (hours)}$$

CAO; condone more than 3sf if correct
 15.40327068 Allow 15h 24m

A1

4

[14]

Q9.

(a) (i) $p \frac{dx}{dt} = q$

where p and q are functions

M1

$$\frac{dx}{dt} = -kx$$

in any correct combination

A1

2

(ii) $-500 = -k \ 20000$ or $500 = k \ 20000$

condone sign error or missing 0

k can be on either side of the equation

M1

$$k = \frac{5}{500} \quad (= 0.025)$$

CSO both (a)(i) and (a)(ii)

A1

2

(b) (i) $A = 1300$

B1

1

(ii) $100 > Ae^{-0.05t}$

condone = for >; condone 99 for 100

M1

$$\ln\left(\frac{100}{A}\right) > -0.05t$$

*take logs correctly
condone 0.5*

m1

$$t > 51.3$$

*or by trial and improvement
(see end of mark scheme)*

A1

population first exceeds 1900 in 2059

F if M1 m1 earned and $t > 0$ following A

A1F

4

ALTERNATIVE METHOD

$$t = 51 \rightarrow 101.5$$

$t = 51$ or $t = 52$ considered

M1

$$t = 52 \rightarrow 96.6$$

$$\Rightarrow 51 < t < 52$$

population first exceeds 1900 in 2059

CAO

A3

4

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[9]

Q10.

- (a) (i) $(5000 - x)$ seen in a product
Could be implied, eg $5000a - xa$

B1

$$\frac{dx}{dt} = kx(5000 - x)$$

B1

2

- (ii) $200 = k \times 1000 \times (5000 - 1000)$

$$\frac{dx}{dt} = 200, x = 1000 \quad \text{in their diff. equation}$$

Condone ts and t = 0 for M1

M1

$$k = 0.00005$$

CAO OE

A1

2

(b) (i)

$$t = 4 \ln \left(\frac{4 \times 2500}{5000 - 2500} \right) = 5.5 \text{ (hours)}$$

$$x \rightarrow 2500 \text{ (or } 4 \ln 4 \text{)}$$

M1

CAO

A1

2

(ii)

$$e^{\frac{30}{4}} = \frac{4x}{5000 - x}$$

OE

B1

M1

$$5000 \times e^{7.5} = x(4 + e^{7.5})$$

Soluble for x

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m1

$$x = 4988.96.. \Rightarrow 4989 \text{ rabbits infected}$$

Or 4988 or 4990; integer value only

A1

4

[10]

Q11.

(a) Using $F = ma$:

$$-4v^{\frac{1}{3}} = 12 \frac{dv}{dt}$$

B1



$$\therefore \frac{dv}{dt} = -\frac{1}{3}v^{\frac{1}{3}}$$

$$-3 \int \frac{dv}{v^{\frac{1}{3}}} = \int dt$$

condone -, 3 incorrect side

M1

$$-3 \times \frac{v^{\frac{2}{3}}}{\frac{2}{3}} = t + c$$

$$-\frac{9}{2}v^{\frac{2}{3}} = t + c$$

condone lack of + c

A1

$$\text{When } t = 0, v = 8 \Rightarrow c = -18$$

M1A1

$$-\frac{9}{2}v^{\frac{2}{3}} = t + 18$$

$$v^{\frac{2}{3}} = 4 - \frac{2}{9}t$$

$$v = \left(4 - \frac{2}{9}t\right)^{\frac{3}{2}}$$

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A1

6

- (b) Particle is at rest when $4 - \frac{2}{9}t = 0$

The value of t is 18

B1

1

[7]

Q12.

- (a) Using $F = ma$

$$1600 \frac{dv}{dt} = 4000 - 40v$$

$$\frac{dv}{dt} = \frac{4000 - 40v}{1600}$$

M1

$$\frac{dv}{dt} = \frac{100 - v}{40}$$

A1

2

(b) $40 \frac{dv}{100 - v} = dt$

B1

$$40 \int \frac{dv}{100 - v} = \int dt$$

M1

$$-40 \ln(100 - v) = t + c$$

Condone lack of '+ c'

A1

When $t = 0$, $v = 0 \Rightarrow c = -40 \ln 100$

$$-40 \ln(100 - v) = t - 40 \ln 100$$

$$t = 40 \ln \frac{100}{100 - v}$$

$$e^{\frac{t}{40}} = \frac{100}{100 - v}$$

M1A1

$$v = 100 - 100 e^{-\frac{t}{40}} \text{ or } 100(1 - e^{-\frac{t}{40}})$$

A1

6

[8]

Q13.

(a) using $F = ma$

$$0.4 \frac{dv}{dt} = 2 - 4v$$

M1

$$\frac{dv}{dt} = -10(v - 0.5)$$

Needs line above

A1

2

(b) hence $\int \frac{1}{v-0.5} dv = - \int 10 dt$
 $\ln(v - 0.5) = -10t + c$

M1 for any side integrated correctly

M1A1

m1 for + c (and M1 gained)

m1

$$v - 0.5 = Ce^{-10t}$$

$$t = 0, v = 1$$

$$\therefore C = 0.5$$

A1

$$\therefore v = 0.5 + 0.5e^{-10t}$$

$$\text{condone } v = 0.5 + e^{-10t-0.693}$$

A1

5

(c) when $v = 0.55$, $0.55 = 0.5 + 0.5e^{-10t}$

substitute 0.55 into C's (b), after finding c, possible numerical error

M1

$$10 = e^{10t}$$

$$t = \ln 10 \div 10$$

A1

$$= 0.230$$

A1

3

[10]

Q14.

- (a) Using $F = ma$:

$$m \frac{dv}{dt} = 49 - 9.8v \quad \text{or} \quad 5g - 9.8v$$

Need to see $m \frac{dv}{dt}$ or $5 \frac{dv}{dt}$ or $a = \frac{49 - 9.81}{5}$

M1

$$\therefore \frac{dv}{dt} = -1.96(v - 5)$$

Must see m terms (not $a = \dots$)

A1

2

(b) $\int \frac{dv}{v-5} = -1.96 \int dt$

And one side integrated

M1

$$\ln(v - 5) = -1.96t + c$$

Need $+c$, A1 each side

A1A1

When $t = 0$, $v = 7 \Rightarrow c = \ln 2$

OE

A1

$$\ln \frac{v-5}{2} = -1.96t$$

$$\frac{v-5}{2} = e^{-1.96t}$$

$$v = 5 + 2e^{-1.96t}$$

CAO

A1

5

[7]

Q15.

- (a) (i) $F = 65g - 260v$
- Accept $260v - 65g$

$$= 65(9.8 - 4v)$$

AG must see 65v or 260

B1

1

(ii) Using $F = ma$

$$65 \frac{dv}{dt} = 65(9.8 - 4v)$$

Need to see terms in m (condone – sign)

M1

$$\frac{dv}{dt} = -4(v - 2.45)$$

AG

A1

2

(b) $\frac{1}{v - 2.45} \frac{dv}{dt} = -4$

$$\int \frac{1}{v - 2.45} dv = -\int 4 dt$$

$$\ln(v - 2.45) = -4t + c$$

M1: log side correct

B1

M1

$$-4t + c$$

A1

$$v - 2.45 = Ce^{-4t}$$

$$t = 0, v = 19.6$$

$$\therefore C = 17.15 \text{ or } e^{2.84}$$

$$\text{Or } c = \ln 17.15 \text{ or } 2.84$$

A1

$$\therefore v = 2.45 + 17.15e^{-4t} \quad 2.45 + 17.2e^{-4t}$$

A1

5

Q16.

(a) Using $F = ma$

$$-2mv^{\frac{5}{4}} = m \frac{dv}{dt}$$

$$\therefore \frac{dv}{dt} = -2v^{\frac{5}{4}} \quad \text{AG}$$

B1: Must see $-2mv^{\frac{5}{4}} = m \frac{dv}{dt}$ or $-2mv^{\frac{5}{4}} = ma$ and correct final answer.

B1

1

(b) $\int \frac{dv}{v^{\frac{5}{4}}} = -2 \int dt$

M1: Two integrals with one in the form $\int f(v)dv$ where $f(v) = v^{\pm \frac{5}{4}}$ or $v^{\pm \frac{1}{5}}$. The other integral must not contain v terms.

M1

$$-\frac{4}{v^{\frac{1}{4}}} = -2t + c$$

A1: Correct expression.

Condone lack of + c for this A1, but no subsequent marks if no c.

A1

When $t = 0$, $v = 16 \Rightarrow c = -2$

dM1: Using $t = 0$ and $v = 16$ to find c .

A1: Obtaining $c = -2$.

dM1

A1

$$-\frac{4}{v^{\frac{1}{4}}} = -2t - 2$$

$$v^{\frac{1}{4}} = \frac{2}{1+t}$$



$$v = \left(\frac{2}{t+1} \right)^4 \quad \text{AG}$$

A1: Correct final answer. Must see $v^{\frac{1}{4}} = \frac{2}{1+t}$

or $v^{-\frac{1}{4}} = \frac{1+t}{2}$ or $\frac{1}{v^{\frac{1}{4}}} = \frac{1+t}{2}$

Or

if they obtain $v = \left(\frac{2}{t+c} \right)^4$

$v = 16, t = 0 \Rightarrow 16^{\frac{1}{4}} = \frac{2}{c}$, condone $c = 1$ (no other root considered)

A1

5

[6]

Q17.

(a) Using $F = ma$,

$$-0.2mv^{\frac{1}{2}} = m \frac{dv}{dt}$$

$$\therefore \frac{dv}{dt} = -0.2v^{\frac{1}{2}}$$

AG Must see equ'n containing m

B1

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(b) $\int \frac{dv}{v^{\frac{1}{2}}} = - \int 0.2 dt$

M1

$$2v^{\frac{1}{2}} = -0.2t + c$$

m1 for $+c$

A1m1

When $t = 0, v = 16 \therefore C = 8$

A1

$$2v^{\frac{1}{2}} = -0.2t + 8$$

$$v = (4 - 0.1t)^2$$

AG

A1

5

- (c) When $v = 1$, $1 = (4 - 0.1t)^2$
 $4 - 0.1t = \pm 1$

M1

$$t = 30 \text{ or } 50$$

$$\left[\begin{array}{l} \text{if use } 2v^{\frac{1}{2}} = 8 - 0.2t \text{ no need to see } 50 \end{array} \right]$$

A1

$$t = 30$$

t ≠ 50 as ball stops when t = 40

A1

3

- (d) Integrating $v = (4 - 0.1t)^2$
 $v = 16 - 0.8t + 0.01t^2$
 $x = 16t - 0.4t^2 + \frac{0.01}{3}t^3 + d$

$-\frac{10}{3}(4 - 0.1t)^3$
M1 for first 3 terms or

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M1

$$\text{When } t = 0, x = 0 \Rightarrow d = 0$$

A1

$$x = 16t - 0.4t^2 + \frac{0.01}{3}t^3$$

$$\text{When speed is } 1 \text{ m s}^{-1}, t = 30$$

$$x = 480 - 360 + 90$$

dep on M1 above

m1

$$= 210$$

[No 'd', 3 marks only]

A1

4

[13]

Q18.

$$\frac{dv}{dt} = -\frac{\lambda}{v^4}$$

M1

$$\int v^{\frac{1}{4}} dv = -\int \lambda dt$$

Condone one of $v^{-\frac{1}{4}}, + \int \lambda dt, \frac{1}{\lambda}$

m1

$$\frac{4}{5} v^{\frac{5}{4}} = -\lambda t + c$$

m1 for + c

A1A1m1

$$t = 0, v = u \therefore c = \frac{4}{5} u^{\frac{5}{4}}$$

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$$\therefore v^{\frac{5}{4}} = u^{\frac{5}{4}} - \frac{5}{4} \lambda t$$

$$v = \left(u^{\frac{5}{4}} - \frac{5}{4} \lambda t \right)^{\frac{4}{5}}$$

A1

[7]

Q19.

- (a) Using $F = ma$:
 $-0.08v^2 = 0.05a$

B1

$$\therefore \frac{dv}{dt} = -1.6v^2$$

AG; condone sign error in first B1

B1

2

(b) $\int \frac{dv}{v^2} = -1.6 \int dt$

M1

$$-\frac{1}{v} = -1.6t(+c)$$

Condone $-\frac{1}{v} = -1.6t + c \Rightarrow \frac{1}{v} = 1.6t + c$

A1

When $t = 0, v = 3 \Rightarrow c = -\frac{1}{3}$

M1

$$\frac{1}{v} = \frac{1}{3} + 1.6t \quad *$$

A1

$$\begin{aligned} \frac{1}{v} &= \frac{1}{3} + \frac{8}{5}t \\ \frac{1}{v} &= \frac{5 + 24t}{15} \\ v &= \frac{15}{5 + 24t} \end{aligned}$$

AG; all working lines correct from *

A1

5

[7]

Q20.

(a) Using $F = ma$:

$$-\lambda m v^{\frac{3}{2}} = m \frac{dv}{dt}$$

M1

$$\therefore \frac{dv}{dt} = -\lambda v^{\frac{3}{2}}$$

AG

A1

2

(b) $\int \frac{dv}{v^{\frac{3}{2}}} = -\lambda \int dt$

M1

$$-\frac{2}{v^{\frac{1}{2}}} = -\lambda t + c$$

Condone no '+c'

A1

When $t = 0, v = 9 \Rightarrow c = -\frac{2}{3}$

Dep. on correct integration

(accept sign or ' $\frac{1}{2}$ ' error)

M1A1

$$\frac{2}{\sqrt{v}} = \lambda t + \frac{2}{3}$$

A1

$$\frac{\sqrt{v}}{2} = \frac{1}{\lambda t + \frac{2}{3}}$$

$$v = \left(\frac{6}{2 + 3\lambda t} \right)^2$$

Needs correct algebra

m1

$$v = \frac{36}{(2 + 3\lambda t)^2}$$

AG

A1

(c) When $v = 4$,

$$\text{or } \frac{36}{(2 + 3\lambda)t^2} = 4 \quad M1$$

$$\frac{2}{\sqrt{v}} = \lambda t + \frac{2}{3} \Rightarrow 1 = \lambda t + \frac{2}{3}$$

$$(2 + 3\lambda t)^2 = 9 \quad A1$$

M1A1

$$t = \frac{1}{3\lambda}$$

$$t = \frac{1}{3\lambda} \quad A1 \text{ needs statement why}$$

$$2 + 3\lambda t \neq -3$$

A1

3

[12]

Q21.

(a) Using $F = ma$

$$-0.05mv = m \frac{dv}{dt}$$

$$\therefore \frac{dv}{dt} = -0.05v$$

Need to see m terms

B1

1

(b) $\int \frac{dv}{v} = -\int 0.05 \, dt$

B1

$$\ln v = -0.05t + c$$

$$v = Ce^{-0.05t}$$

Need first 2 terms

M1

When $t = 0$, $v = 20$

$$\therefore C = 20$$

$$v = 20e^{-0.05t}$$

only for fully correct solutions

M1A1

4

(c) When $v = 10$, $10 = 20e^{-0.05t}$

M1

$$e^{0.05t} = 2$$

A1

$$\therefore t = \frac{1}{0.05} \ln 2$$

$$= 13.9 \text{ or } 20 \ln 2$$

A1

3

(d) Work done by force is change in KE of car

M1

$$= \frac{1}{2} m (20)^2 - \frac{1}{2} m (10)^2$$

A1

$$= 150 m$$

A1

3

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[11]

Q22.

(a) Using $F = ma$

$$-0.05mv = m \frac{dv}{dt}$$

$$\therefore \frac{dv}{dt} = -0.05v$$

Need to see m terms

B1

1

(b) $\int \frac{dv}{v} = -\int 0.05 dt$

B1

$$\ln v = -0.05t + c$$

$$v = Ce^{-0.05t}$$

Need first 2 terms

M1

$$\text{When } t = 0, v = 20$$

$$\therefore C = 20$$

$$v = 20e^{-0.05t}$$

fully correct solutions

M1A1

4

(c) When $v = 10$, $10 = 20e^{-0.05t}$

M1

$$e^{0.05t} = 2$$

A1

$$\therefore t = \frac{1}{0.05} \ln 2$$

$$= 13.9$$

Accept 20 ln 2

A1

3

[8]

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Q23.

(a) Using $F = ma$:

$$-\lambda mv = ma = m \frac{dv}{dt}$$

Condone no '-'

M1

$$\therefore \frac{dv}{dt} = -\lambda v$$

AG

Note: no use of $m \Rightarrow$ no marks in (a)

A1

2

(b) $\int \frac{dv}{v} = -\lambda \int dt$

M1

$$\ln v = -\lambda t + c$$

$$v = C e^{-\lambda t}$$

Needs '+ c'

A1

When $t = 0$, $v = U \Rightarrow C = U$

Needs correct working

M1

$$v = U e^{-\lambda t}$$

AG

A1

4

[6]

Q24.

$$1600 \frac{dv}{dt} = -40v$$

Applying Newton's second law with $40v$ and $\frac{dv}{dt}$

M1

Correct equation

EXAM PAPERS PRACTICE

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$$\int \frac{1}{v} dv = \int -\frac{1}{40} dt$$

Separating variables

dM1

$$\ln v = -\frac{t}{40} + c$$

integrating to get $\ln v$ term.

dM1

$$v = A e^{-\frac{t}{40}}$$

Correct integral with or without c

A1

$$t = 0, v = 20 \Rightarrow c = 20$$

Finding constant

dM1

$$v = 20e^{-\frac{t}{40}}$$

Correct final result

A1

[7]

Q25.

- (a) The boat is not affected by the movement of the water.

First assumption

B1

The resistance force will directly oppose the motion of the boat and be the only force that needs to be considered.

second assumption

B1

2

- (b) The boat is a particle. / There is no wind. / No air resistance.

Appropriate assumption

B1

1

(c) (i) $80 \frac{dv}{dt} = -20v$

Use of Newton's second law, $\frac{dv}{dt}$ and $20v$.

M1

$$\frac{dv}{dt} = -\frac{v}{4}$$

Correct result

A1

2

(ii) $\int \frac{1}{v} dv = -\int \frac{1}{4} dt$

Sep. of variables and forming integrals

M1

$$\ln v = -\frac{t}{4} + c$$

Integrating to get an $\ln v$ term

dM1

Correct integrals with or without c

A1

$$v = Ae^{-\frac{t}{4}}$$

$$v = 12, t = 0 \Rightarrow A = 12$$

Finding A or c

dM1

$$v = Ae^{-\frac{t}{4}}$$

Correct final result

A1

5

[10]

Q26.

(a) $20 \frac{dv}{dt} = -10\sqrt{v}$

applying Newton's second law with $\frac{dv}{dt}$

M1

correct differential equation

A1

$$\frac{dv}{dt} = -\frac{\sqrt{v}}{2}$$

separating variables

dM1

$$\int \frac{1}{\sqrt{v}} dv = \int -\frac{1}{2} dt$$

AG

$$2\sqrt{v} = \frac{t}{2} + c$$

integrating

dM1

correct integrals

A1

$$t = 0, v = 25 \Rightarrow c = 10$$

finding the constant of integration

$$v = \left(\frac{20 - t}{4} \right)^2$$

correct final result from correct working

A1

7

(b) $t = 20$

correct time

B1

1

[8]

Q27.

$$\frac{dv}{dt} = \frac{k}{v}$$

B1

$$\int v dv = \int k dt$$

Separation of variables involving t

M1

$$\frac{v^2}{2} = kt + c$$

m1

Integrate

A1

$$t = 0, v = u, \therefore c = \frac{u^2}{2}$$

m1

$$v^2 = u^2 + 2kt$$

A1

[6]

Q28.

(a) $mg - 0.1m v = m \frac{dv}{dt}$
Newton's 2nd law

M1

$$\frac{dv}{dt} = g - 0.1v$$

CAO

A1

2

(b) $\int_0^v \frac{dv}{g - 0.1v} = \int_0^t dt$
Attempt at integration with correct separation of variables

M1

$$[-10 \ln(g - 0.1v)]_0^v = [t]_0^t$$

-1 EE

A2,1

$$\ln \frac{g - 0.1v}{g} = -0.1t$$

A1F

$$v = 10g(1 - e^{-0.1t})$$

Or equivalent

A1F

5

(c) As $t \rightarrow \infty$, $e^{-0.1t} \rightarrow 0$

M1

1

$$\therefore v \rightarrow 10g = 98$$

[8]

Q29.

(a) $1000 \frac{dv}{dt} = 2000 - 40v$

Use of Newton's 2nd law

M1

$$\frac{dv}{dt} = \frac{2000 - 40v}{1000} = \frac{50 - v}{25}$$

Correct expression from correct working

A1

2

(b) $\int \frac{1}{50 - v} dv = \int \frac{1}{25} dt$

$$-\ln|50 - v| = \frac{t}{25} + c$$

Integration to obtain t and 1n term

M1

Correct integration

A1

$$50 - v = Ae^{\frac{t}{25}}$$

$$v = 50 - Ae^{-\frac{t}{25}}$$

Solving for v

m1

$$v = 0, t = 0 \Rightarrow A = 50$$

Finding constant of integration.

m1

$$v = 50 \left(1 - e^{-\frac{t}{25}} \right)$$

Correct final answer

A1

5

[7]

Q30.

(a) $m \frac{dv}{dt} = -mkv^3$

Forming a differential equation

M1

$$\int \frac{1}{v^3} dv = -\int k dt$$

$$-\frac{1}{2v^2} = -kt + c$$

Integrating to get a $\frac{1}{v^2}$ term
Correct Integral including c

m1 A1

$$v = U, t = 0 \Rightarrow c = -\frac{1}{2U^2}$$

Finding c
Correct c

m1 A1

$$\frac{1}{2v^2} = kt + \frac{1}{2U^2} = \frac{2ktU^2 + 1}{2U^2}$$

$$v^2 = \frac{U^2}{2ktU^2 + 1}$$

Correct final answer from correct working

A1

6

(b) v tends to zero

Allow decreases

B1

1

[7]